

Name: _____

UNIT 30: MOLECULAR GENETICS

Topic 2: Transcription & Translation

Gene Expression

Recall that DNA provides the “instructions” for each of our cells. Each segment of DNA responsible for encoding a specific protein, which carries out a specific function. Not every gene, however, is expressed in each cell. This is what allows cells to specialize.

➤ *Some genes do not appear to code for any proteins. Does this DNA have a purpose?*

➤ *Outline the “central dogma” of gene expression.*

➤ *List three differences between DNA and RNA in the table below.*

DNA	RNA

Name: _____

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Transcription & Translation

Complete the following table comparing the processes of transcription and translation.

	Transcription	Translation
Where it occurs		
Main purpose		
Steps involved		

➤ *Explain how mRNA is modified prior to translation.*

➤ *Outline the role of ribosomes in translation.*

Name: _____

- *Example: Transcribe the following sequence of DNA into mRNA, and then translate the strand of mRNA into a sequence of amino acids...*

5' TTCAGACCGATCTTA 3'
3' AAGTCTGGCTAGAAT 5'

- *What is a codon? Distinguish between stop and start codons.*

- *What is an anticodon?*

Name: _____

PRACTICE #1

- Transcribe the following strand of DNA into mRNA:

C C T T A C T T A T A A T G C T C A T G C T A
G G A A T G A A T A T T A C G A G T A C G A T

- Translate your mRNA strand above into a sequence of amino acids:

PRACTICE #2

- How many nucleotides are required to code for the following sequence of amino acids?

Leu - Tyr - Arg - Trp - Ser

- Is it possible to determine the mRNA sequence that is responsible for producing the following sequence of amino acids? Explain...
- What does this illustrate?

Name: _____

PRACTICE #3

- Identify a single-letter mutation that could occur in the following DNA sequence to produce a stop codon at some point in the amino acid:

AACGAATATCGG

RE-CAP: TYPES OF NUCLEIC ACIDS

Nucleic Acid	Structure	Function
DNA		
mRNA		
tRNA		
rRNA		

Name: _____